

MASTER'S COMPREHENSIVE EXAMINATION

Part I

Theory (90 minutes)

December 4, 2010

Instructions: Attempt any three out of the following four questions. Mention which three you would like to have graded. Open book and open notes.

- 1) A bowl on my desk has 5 cashew nuts, 5 walnuts, and 5 peanuts. I pick nuts at random from the bowl, one at a time, and eat them as I pick them.
 - a) Given that I have eaten exactly 5 nuts, what is the expected number of cashews left in the bowl?
 - b) Given I have eaten exactly 5 nuts, what is the probability that I have eaten at least one of each type of nut?
 - c) Given I have eaten exactly 5 nuts, what is the expected number of different types of nuts I have eaten?

- 2) Let the waiting time until a light bulb fails be exponential, with expected waiting time of 1000 hours. Assume that the lifetimes of light bulbs are independent of each other. Four new lightbulbs are put in a room, and all four lights are turned on and kept on.
 - a) What is the expected waiting time until at least 2 of the 4 light bulbs fail?
 - b) What is the expected number of light bulbs that fail in the first 1000 hours?
 - c) What is the probability that in the first 2000 hours that at least 3 of the light bulbs fail?

- 3) (a) Results of my last 8 tennis matches were as follows: win, lose, lose, lose, lose, lose, lose, lose. Find the exact P -value using a sign test to determine if my probability of winning is the same as my probability of losing vs the alternative that these probabilities are different.

(b) Find the approximate P -value that my probability to win is the same as that to lose if I won only 10 matches out of my last 100.

- 4) Let Y_1, Y_2, Y_3 be a random sample of size 3 from a distribution with density

$$f_Y(y) = (\theta + 1)y^\theta, \quad 0 < y < 1, \theta > -1.$$

- a) Find a one-dimensional sufficient statistics.
- b) Find the MLE of θ .